Worked for 12s

Azure Firewall Complete Practical – NAT, Application an…

# 🔥 Azure Firewall Complete Practical

### VM → AzureFirewallSubnet → Firewall → DNAT → UDR → Application Rules → Network Rules

> **Important correction:** Azure Firewall me **smaller priority number = higher priority** hoti hai. Example: Priority `300` ka rule, Priority `400` se pehle process hoga. Classic rules me overall processing order **DNAT** → **Network** → **Application** hota hai.

---

# 🇮🇳 PART 1 — ROMAN HINDI

## 🏗️ Step 1 — Windows VM Create Kiya

Sabse pehle humne normal Azure VM ki tarah ek **Windows Server 64-bit VM** create kiya.

### VM configuration

- Resource Group select/create kiya.
- Windows Server image select ki.
- VM Size select ki.
- Username aur Password configure kiya.
- VNet select/create kiya.
- VM ke liye normal workload subnet use kiya.
- RDP port `3389` required tha.
- VM deploy kar diya.

### 🎯 Purpose

Ye VM hamara **testing machine** tha.

Isi VM ko humne baad me:

- RDP se access kiya.
- Internet traffic test kiya.

- Google block test kiya.
- Facebook test kiya.
- ChatGPT test kiya.
- Azure Portal test kiya.
- Network rules test kiye.

---

# 🌐 Step 2 — Azure Firewall Ke Liye Dedicated Subnet Create Kiya

VM create hone ke baad:

**Azure Portal → Virtual Networks → Our VNet → Subnets**

gaye.

Phir:

**+ Subnet**

select kiya.

Firewall ke liye dedicated subnet create kiya:

`AzureFirewallSubnet`

⚠️ Is subnet ka naam exactly **AzureFirewallSubnet** hona chahiye.

Is subnet ke andar normal VM deploy nahi kiya.

## Architecture

```
Virtual Network
|
├── VM-Subnet
|       └── Windows VM
|
└── AzureFirewallSubnet
        └── Azure Firewall
```

---

# 🔥 Step 3 — Azure Firewall Create Kiya

Azure Portal ke search bar me search kiya:

**Firewall**

Then:

**Firewalls → + Create**

Firewall ka naam diya:

`Firewall-Testing-01`

### Configuration

```
Firewall Name : Firewall-Testing-01
Region        : Same region as VNet
Virtual Network: Existing VNet
Firewall Subnet: AzureFirewallSubnet
```

Firewall management ke liye humne:

**Classic Firewall Rules**

use kiye.

Microsoft ab generally Firewall Policy recommend karta hai, lekin hamare practical me humne **Rules (classic)** use kiya.

---

## 🌍 Step 4 — Firewall Public IP Create Kiya

Firewall creation ke time:

**Public IP Address → Add new**

select kiya.

Example:

`Firewall-PIP-01`

Public IP create hone ke baad firewall ke paas do important IPs aa gaye:

```
Firewall Public IP
Firewall Private IP
```

### In dono ka use

**Public IP**

Internet se Firewall tak traffic lane ke liye.

**Private IP**

Internal VM traffic ko UDR ke through Firewall tak bhejne ke liye.

---

## ✅ Step 5 — Review + Create

Configuration verify ki.

Then:

**Review + Create → Create**

Firewall successfully deploy ho gaya.

Deployment complete hone ke baad humne firewall ke:

- Public IP
- Private IP

note kar liye.

---

## 🖥️ Step 6 — Direct VM RDP Test Kiya

Ab VM ko RDP karne ki koshish ki.

Lekin humara objective tha:

```
Internet
   ↓
Firewall Public IP
   ↓
Azure Firewall
   ↓
VM Private IP
```

Iske liye normal direct VM public IP nahi, balki **Firewall DNAT Rule** use karna tha.

---

## 🟠 PART A — NAT / DNAT RULE PRACTICAL

## 🔀 Step 7 — NAT Rule Collection Open Kiya

Azure Portal me:

**Azure Firewall → Rules (classic)**

gaye.

Yahan hume teen rule types mile:

```
1. NAT Rule Collection
2. Network Rule Collection
3. Application Rule Collection
```

Sabse pehle:

**NAT Rule Collection**

open kiya.

---

## 🔀 Step 8 — NAT Collection Create Kiya

Click:

**+ Add NAT Rule Collection**

Example:

```
Collection Name : NAT-Collection-01
Priority        : 200
Action          : DNAT
```

DNAT ka matlab:

**Destination Network Address Translation**

Azure Firewall DNAT inbound traffic ko Firewall Public IP se internal resource ke Private IP tak translate karta hai.

---

## 🖥️ Step 9 — RDP DNAT Rule Create Kiya

Rule Name:

`NAT-RDP-Rule-01`

Configuration:

```
Protocol            : TCP

Source Type         : IP Address
Source              : *

Destination Address : Firewall Public IP

Destination Port    : 3389

Translated Address  : VM Private IP

Translated Port     : 3389
```

Then:

**Add / Save**

---

## 🔄 NAT Traffic Flow

Ab jab hum:

```
mstsc
```

open karke Firewall Public IP use karte hain:

```
My Laptop
   |
   | TCP 3389
   ↓
Firewall Public IP
   |
   | DNAT
   ↓
VM Private IP : 3389
   |
   ↓
Windows VM
```

Azure Firewall destination ko translate karta hai:

```
Firewall-Public-IP:3389
          ↓
```

```
VM-Private-IP:3389
```

Microsoft bhi DNAT ko inbound public-to-private translation ke liye use karta hai.

---

## ✅ Step 10 — RDP Successful

NAT rule save hone ke baad dobara RDP kiya.

Is baar VM successfully access ho gaya.

### Result

```
Before DNAT ❌
Firewall Public IP → VM access nahi

After DNAT ✅
Firewall Public IP → VM Private IP → RDP
```

---

## 🛣️ PART B — ROUTE TABLE / UDR

Ab hume VM ka **outbound traffic bhi Firewall se bhejna tha**.

Example:

```
VM → Google
VM → Facebook
VM → ChatGPT
```

Agar route table nahi banate, to VM ka internet traffic Firewall ko bypass kar sakta hai.

Isliye humne **UDR – User Defined Route** create kiya.

---

## 🛣️ Step 11 — Route Table Create Kiya

Azure Portal search:

**Route Tables**

Then:

**+ Create**

Name:

`Route-Table-01`

Create kar diya.

---

## ➕ Step 12 — Default Route Add Kiya

Route Table open kiya.

Left side:

**Routes → + Add**

Configuration:

```
Route Name:
Route-01

Destination Type:
IP Addresses

Destination IP/CIDR:
0.0.0.0/0

Next Hop Type:
Virtual Appliance

Next Hop Address:
Azure Firewall Private IP
```

⚠️ Yahan **VM ki IP nahi**, Azure Firewall ki **Private IP** deni hai.

Microsoft ke Azure Firewall routing example me bhi default route `0.0.0.0/0` ka next hop Firewall private IP rakha jata hai.

---

## 🌍 0.0.0.0/0 Ka Meaning

```
0.0.0.0/0
```

ka simple meaning:

**All destination IPv4 traffic**

Yani VM kisi bhi Internet destination ko access kare:

```
Google
Facebook
GitHub
Microsoft
ChatGPT
Any website
```

traffic ka next hop hoga:

**Azure Firewall**

---

## 🔄 Traffic Flow

```
Windows VM
    ↓
VM Subnet
    ↓
UDR
0.0.0.0/0
    ↓
Next Hop = Virtual Appliance
    ↓
Azure Firewall Private IP
    ↓
Firewall Rules
    ↓
Internet
```

---

## ❌ Step 13 — Route Create Karne Ke Baad Bhi Rule Work Nahi Hua

Humne VM me browser open kiya.

Google ab bhi open ho raha tha.

Reason:

**Route Table bana hua tha, lekin VM wale subnet ke saath associate nahi tha.**

Yani:

```
Route exists ✅

But

Subnet association ❌
```

Is wajah se VM us route ko use hi nahi kar raha tha.

---

## 🔗 Step 14 — Route Table Ko VM Subnet Ke Saath Associate Kiya

Route Table:

`Route-Table-01`

open kiya.

Then:

**Subnets → + Associate**

Select kiya:

```
Virtual Network:
Our VNet

Subnet:
VM Subnet
```

Then:

**OK / Add**

---

## ✅ Ab Real Traffic Flow Bana

```
Windows VM
     ↓
VM Subnet
     ↓
Route Table
     ↓
0.0.0.0/0
```

```
        ↓
Azure Firewall Private IP
        ↓
Firewall
        ↓
Rules
        ↓
Internet
```

Ab Firewall traffic path me aa gaya.

---

## 🟢 PART C — APPLICATION RULE ALLOW PRACTICAL

Application Rules Layer 7 par mainly:

```
HTTP
HTTPS
FQDN
```

ko control karti hain.

---

## 🌐 Step 15 — Application Allow Collection Create Kiya

Path:

**Azure Firewall → Rules (classic) → Application Rule Collection**

Click:

**+ Add Application Rule Collection**

Example:

```
Collection Name:
App-Allow-Collection-01

Priority:
400
```

```
Action:
Allow
```

---

## 🟢 Step 16 — Allow All Web Rule Create Kiya

Rule Name:

`App-Allow-Rule-01`

Configuration:

```
Source Type:
IP Address

Source:
VM Private IP
ya
VM Subnet CIDR

Protocols:
HTTP:80
HTTPS:443

Target FQDN:
*
```

⚠️ Source me generally **VM ki Private IP ya VM subnet range** dena correct hai.

Firewall Private IP ko source dena nahi chahiye agar traffic VM se originate ho raha hai.

Target:

```
*
```

ka meaning:

**Any FQDN / website matching this rule.**

Save kar diya.

---

# 🔴 PART D — APPLICATION RULE DENY PRACTICAL

Ab requirement:

```
Facebook     ✅ Allow
ChatGPT      ✅ Allow
Azure Portal ✅ Allow
Google       ❌ Block
```

---

## 🚫 Step 17 — Google Deny Collection Create Kiya

Again:

**Application Rule Collection → Add**

Collection:

```
Name:
App-Deny-Collection-01

Priority:
300

Action:
Deny
```

Allow collection:

```
Priority 400
```

Deny collection:

```
Priority 300
```

Because:

```
Lower Number = Higher Priority
```

Isliye:

```
300 → Process First
400 → Process Later
```

## 🚫 Step 18 — Google Deny Rule Create Kiya

Rule:

`App-Deny-Google-01`

Configuration:

```
Source Type:
IP Address

Source:
VM Private IP / VM Subnet

Protocols:
HTTP:80
HTTPS:443

Target FQDN:
google.com
*.google.com
```

⚠️ Sirf:

```
*.google.com
```

subdomains ko match karta hai.

Reliable test ke liye:

```
google.com
*.google.com
```

dono include karna better hai.

---

## 🔄 Application Rule Processing

Google request:

```
VM
 ↓
UDR
 ↓
```

```
Firewall
 ↓
App-Deny-Collection-01
Priority 300
 ↓
google.com MATCH
 ↓
DENY ❌
```

Facebook request:

```
VM
 ↓
Firewall
 ↓
Deny Google Rule
 ↓
No Match
 ↓
Allow Collection
Priority 400
 ↓
Target *
 ↓
ALLOW ✅
```

---

# 🧪 Step 19 — Browser Testing

VM ke andar browser open kiya.

Test:

```
facebook.com
✅ Open

chatgpt.com
✅ Open

portal.azure.com
✅ Open

google.com
❌ Block
```

Iska matlab Application Rule successfully work kar raha tha.

---

## 🔵 PART E — NETWORK RULE PRACTICAL

Network Rules Layer 3 / Layer 4 basis par traffic control karti hain.

Mainly:

```
Source IP
Destination IP
Port
Protocol
```

ke according filtering hoti hai.

---

## 🟢 Step 20 — Network Allow Rule Collection

Path:

**Azure Firewall → Rules (classic) → Network Rule Collection**

Click:

**+ Add Network Rule Collection**

Example:

```
Collection Name:
Network-Allow-Collection-01

Priority:
500

Action:
Allow
```

Rule:

```
Rule Name:
Network-Allow-HTTPS-01

Protocol:
TCP
```

```
Source:
VM Private IP / VM Subnet

Destination:
Required Destination IP

Destination Port:
443
```

Save kiya.

### Meaning

```
VM
 ↓
TCP 443
 ↓
Destination IP
 ↓
ALLOW
```

---

## 🔴 Step 21 — Network Deny Rule Collection

Ek second Network Rule Collection create ki.

Example:

```
Collection Name:
Network-Deny-Collection-01

Priority:
450

Action:
Deny
```

Priority `450`, Allow ke `500` se smaller hai.

Isliye deny pehle evaluate hoga.

Rule example:

```
Rule Name:
Network-Deny-Rule-01

Protocol:
TCP

Source:
VM Private IP / VM Subnet

Destination:
Specific Destination IP

Destination Port:
443
```

Save.

---

## 🔄 Network Rule Result

Blocked destination:

```
VM
 ↓
Network Deny
Priority 450
 ↓
Destination matches
 ↓
DENY ❌
```

Other allowed destination:

```
VM
 ↓
Deny Rule
No Match
 ↓
Allow Rule
Priority 500
 ↓
ALLOW ✅
```

---

# 🧠 NAT vs Network vs Application — Practical Understanding

| Rule | Main Purpose | Hamara Practical |
|---|---|---|
| NAT / DNAT | Internet se internal VM access | Firewall Public IP `3389` → VM Private IP `3389` |
| Network Rule | IP + Port + Protocol filtering | TCP/IP/Port traffic Allow/Deny |
| Application Rule | Website/FQDN filtering | Google Deny, baaki websites Allow |

---

## 🔥 Azure Firewall Rule Processing Order

Azure Firewall ka important processing order:

```
1️⃣ DNAT Rules
        ↓
2️⃣ Network Rules
        ↓
3️⃣ Application Rules
        ↓
4️⃣ No match = Deny
```

Network Rule match ho gaya to Application Rule tak traffic normally nahi jata because firewall rules are terminating.

---

## 🏗️ Final Architecture

```
                    INTERNET
                       │
          ┌────────────┴────────────┐
          │                         │
      INBOUND                   OUTBOUND
        RDP                       WEB
          │                         │
          ▼                         ▲
   Firewall Public IP               │
          │                         │
```

```
          DNAT Rule                  │
              │                Azure Firewall
              ▼                      ▲
        VM Private IP                │
          TCP 3389                   │
              │                      │
              ▼                      │
         Windows VM ─── VM Subnet ─── Route Table
                                  0.0.0.0/0
                                     │
                                     ▼
                           Next Hop: Virtual Appliance
                                     │
                                     ▼
                           Firewall Private IP
                                     │
                      ┌──────────────┴──────────────┐
                      │                             │
                Network Rules               Application Rules
                IP/Port/Protocol                FQDN/HTTP/S
                      │                             │
                      └──────────────┬──────────────┘
                                     ▼
                                  Internet
```

---

## 🧪 Final Practical Verification

### Test 1 — RDP

```
My Laptop
   ↓
Firewall Public IP:3389
   ↓
DNAT
   ↓
VM Private IP:3389

Result: ✅ Successful
```

### Test 2 — Facebook

```
VM → UDR → Firewall → App Rules → Facebook
```

```
Result: ✅ Allowed
```

### Test 3 — ChatGPT

```
VM → UDR → Firewall → App Rules → ChatGPT
```

```
Result: ✅ Allowed
```

### Test 4 — Azure Portal

```
VM → UDR → Firewall → App Rules → Azure Portal
```

```
Result: ✅ Allowed
```

### Test 5 — Google

```
VM → UDR → Firewall
       ↓
Google Deny Rule
       ↓
DENY
```

```
Result: ❌ Blocked
```

---

# 🇬🇧 PART 2 — ENGLISH

## 🔥 Azure Firewall Practical — Complete Step-by-Step Flow

### Step 1 — Create the Windows VM

First, we created a normal **Windows Server 64-bit Azure VM**.

The VM was our test machine for:

- RDP connectivity.
- Internet connectivity.
- Application rules.
- Network rules.
- Google blocking.
- General website access.

---

## Step 2 — Create Azure Firewall Subnet

We opened:

**Virtual Network** → **Subnets** → **+ Subnet**

and created the dedicated subnet:

`AzureFirewallSubnet`

The architecture became:

```
Virtual Network
|
├── VM Subnet
|     └── Windows VM
|
└── AzureFirewallSubnet
      └── Azure Firewall
```

---

## Step 3 — Create Azure Firewall

We searched:

**Azure Portal** → **Firewall** → **Create**

Firewall name:

`Firewall-Testing-01`

We selected our existing VNet and deployed the Firewall into:

`AzureFirewallSubnet`

For this practical, we used:

**Firewall Rules (Classic)**

instead of a Firewall Policy.

---

## Step 4 — Create Firewall Public IP

During Firewall creation, we selected:

**Public IP → Add New**

Example:

`Firewall-PIP-01`

After deployment, the Firewall had:

```
Public IP
Private IP
```

The Public IP was used for inbound traffic.

The Private IP was later used as the next hop in our Route Table.

---

# 🟠 NAT / DNAT PRACTICAL

## Step 5 — Open NAT Rule Collection

We opened:

**Azure Firewall → Rules (classic) → NAT Rule Collection**

and selected:

**Add NAT Rule Collection**

Collection:

```
Name     : NAT-Collection-01
Priority : 200
Action   : DNAT
```

---

## Step 6 — Create the RDP DNAT Rule

Rule configuration:

```
Rule Name           : NAT-RDP-Rule-01
Protocol            : TCP
Source              : *
Destination Address : Firewall Public IP
Destination Port    : 3389
Translated Address  : VM Private IP
Translated Port     : 3389
```

After saving the rule, the traffic flow became:

```
Internet
   ↓
Firewall Public IP:3389
   ↓
Azure Firewall DNAT
   ↓
VM Private IP:3389
   ↓
Windows VM
```

Azure Firewall DNAT translates the firewall-facing destination into the internal private destination.

---

### Step 7 — Test RDP

We connected to:

```
Firewall Public IP
```

using Remote Desktop.

The Firewall translated the request to:

```
VM Private IP:3389
```

Result:

**RDP successful. ✅**

---

## 🛣️ ROUTE TABLE / UDR

### Step 8 — Create Route Table

Next, we needed the VM's outbound traffic to pass through Azure Firewall.

We created:

`Route-Table-01`

---

### Step 9 — Add Default Route

We opened:

**Route Table → Routes → Add**

and configured:

```
Route Name:
Route-01

Destination Type:
IP Addresses

Destination:
0.0.0.0/0

Next Hop:
Virtual Appliance

Next Hop Address:
Azure Firewall Private IP
```

`0.0.0.0/0` represents the default IPv4 route.

Therefore, Internet-bound VM traffic is forwarded toward Azure Firewall.

---

## Step 10 — Initial Test Failed

We tested from the VM, but Internet traffic was still not behaving according to our Firewall rules.

The reason was:

**The Route Table existed, but it was not associated with the VM subnet.**

---

## Step 11 — Associate Route Table with VM Subnet

We opened:

**Route Table → Subnets → Associate**

and selected:

```
Virtual Network:
Our VNet

Subnet:
VM Subnet
```

After the association, the traffic path became:

```
VM
 ↓
VM Subnet
 ↓
Route Table
 ↓
0.0.0.0/0
 ↓
Virtual Appliance
 ↓
Azure Firewall Private IP
 ↓
Firewall Rules
 ↓
Internet
```

---

# 🟢 APPLICATION ALLOW RULE

## Step 12 — Create Allow Collection

We opened:

**Azure Firewall → Rules (classic) → Application Rule Collection**

and created:

```
Name:
App-Allow-Collection-01

Priority:
400

Action:
Allow
```

Rule:

```
Name:
App-Allow-Rule-01

Source:
VM Private IP / VM Subnet

Protocols:
HTTP:80
HTTPS:443

Target FQDN:
*
```

This allowed general HTTP/HTTPS destinations.

Application rules are Layer 7 rules used to filter traffic based on FQDNs and HTTP/HTTPS application information.

---

# 🔴 APPLICATION DENY RULE

## Step 13 — Create Google Deny Collection

We created another Application Rule Collection:

```
Name:
App-Deny-Collection-01

Priority:
300

Action:
Deny
```

Priority `300` is processed before `400` because a smaller priority number represents a higher priority.

Rule:

```
Name:
App-Deny-Google-01

Source:
VM Private IP / VM Subnet
```

```
Protocols:
HTTP:80
HTTPS:443

Target FQDN:
google.com
*.google.com
```

---

### Step 14 — Test Application Rules

From the Windows VM:

```
facebook.com
✅ Open

chatgpt.com
✅ Open

portal.azure.com
✅ Open

google.com
❌ Blocked
```

This confirmed that the VM traffic was now passing through Azure Firewall and the Application Rules were being evaluated.

---

## 🔵 NETWORK RULE PRACTICAL

### Step 15 — Create Network Allow Collection

We opened:

**Azure Firewall → Rules (classic) → Network Rule Collection**

and created:

```
Name:
Network-Allow-Collection-01

Priority:
```

```
500

Action:
Allow
```

Example rule:

```
Name:
Network-Allow-HTTPS-01

Protocol:
TCP

Source:
VM Private IP / VM Subnet

Destination:
Required Destination IP

Destination Port:
443
```

Result:

```
VM → Destination IP:443
✅ Allowed
```

Network Rules work primarily with IP addresses, ports, and protocols at Layers 3 and 4.

---

# 🔴 NETWORK DENY RULE

## Step 16 — Create Network Deny Collection

We created another collection:

```
Name:
Network-Deny-Collection-01

Priority:
450

Action:
Deny
```

Example rule:

```
Rule Name:
Network-Deny-Rule-01

Protocol:
TCP

Source:
VM Private IP / VM Subnet

Destination:
Specific Destination IP

Destination Port:
443
```

Because Priority `450` is evaluated before Priority `500`, matching traffic is denied first.

---

# 🧠 Final Understanding

## NAT Rule

```
Internet
 ↓
Firewall Public IP
 ↓
DNAT
 ↓
Private VM
```

Used mainly for inbound traffic.

Our practical:

```
Firewall Public IP:3389
          ↓
VM Private IP:3389
```

---

## Network Rule

```
Source IP
+
Destination IP
+
Protocol
+
Port
```

Used for Layer 3/Layer 4 traffic control.

Example:

```
VM → Destination IP → TCP 443
```

Allow or Deny.

---

### Application Rule

```
Source
+
HTTP/HTTPS
+
FQDN
```

Used mainly for Layer 7 web filtering.

Our practical:

```
Facebook       → Allow ✅
ChatGPT        → Allow ✅
Azure Portal   → Allow ✅
Google         → Deny ❌
```

---

## 🔥 Final Azure Firewall Rule Order

The important rule-processing order is:

```
1. DNAT Rules
       ↓
2. Network Rules
       ↓
3. Application Rules
```

```
       ↓
4. Default Deny
```

A Network Rule match terminates processing, so Application Rules are not evaluated for that connection afterward.

---

## 🎯 Complete Practical in One Line

```
Create VM
   ↓
Create AzureFirewallSubnet
   ↓
Create Azure Firewall
   ↓
Create Firewall Public IP
   ↓
Create DNAT Rule for RDP
   ↓
RDP to VM through Firewall
   ↓
Create Route Table
   ↓
0.0.0.0/0
   ↓
Next Hop = Azure Firewall Private IP
   ↓
Associate Route Table with VM Subnet
   ↓
Create Network Allow/Deny Rules
   ↓
Create Application Allow Rule
   ↓
Create Higher-Priority Google Deny Rule
   ↓
Test from VM
   ↓
Facebook ✅
ChatGPT ✅
Azure Portal ✅
Google ❌
RDP ✅
```

## ✅ Final Result

Humne successfully Azure Firewall ke through **inbound aur outbound dono traffic control** kiya.

**Inbound side:**

```
NAT/DNAT → Firewall Public IP → VM Private IP → RDP
```

**Outbound side:**

```
VM → UDR → Azure Firewall → Network/Application Rules → Internet
```

Aur isi practical me humne Azure Firewall ke tino major rule types practically cover kiye:

**NAT Rule ✅ | Network Rule ✅ | Application Rule ✅**